

Zhouhao Yang

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PROFILE OVERVIEW

- 3rd year Ph.D. candidate in Applied Mathematics and Statistics at Johns Hopkins University, specializing in stochastic control, mean field games, reinforcement learning, and diffusion models.
- First-author research in generative modeling for heavy-tailed data, stochastic control and financial applications, and distributionally robust learning, combining mathematical guarantees with large-scale empirical evaluation.

EDUCATION

- Johns Hopkins University** Aug 2024 – Present
Ph.D. Candidate, Department of Applied Mathematics and Statistics
Advisor: Haoyang Cao
Research Direction: Reinforcement Learning, Diffusion Models, Stochastic Control Theory, Mean Field Games.
Selected Coursework: Stochastic Control, Games, and Learning (A), Data Science Method for Large Scale Graphs (A), Nonlinear Optimization (A), Statistical Theory (A), Random Matrix Theory in Data Science and Statistics (A).
- National University of Singapore** Aug 2023 – Jun 2024
Ph.D. Candidate, Department of Computer Science (Transferred to Johns Hopkins)
Advisor: Kenji Kawaguchi and Johnathan Scarlett
Research Direction: AI for Science, Physics-Informed Neural Networks, Memory-Efficient Finetuning for LLMs.
- Shanghai Jiao Tong University** Sep 2019 – Jun 2023
Bachelor of Mathematics and Applied Mathematics (Zhiyuan Honors Program)
Advisor: Dan Hu

RESEARCH EXPERIENCE (SELECTED)

- JHU: Heavy-tailed Flow Matching via Random Clocks** March 2026-May 2026
- Advisor(s): Haoyang Cao and Vladimir Braverman.
 - Studied the generation & prediction of heavy-tailed data via flow matching and diffusion models, especially useful in fields like finance and weather forecast.
 - Developed a unified flow-matching framework using path-valued random clocks to generally model Gaussian, α -stable, and Student-t source distributions while preserving tractable conditional-Gaussian training objectives.
 - Encoded random-clock paths with truncated logsignature features for future training.
 - Evaluated the method on imbalanced α -stable mixtures, CIFAR10-LT, and 128×128 HRRR weather fields. Our method outperformed heavy-tailed diffusion and ODE baselines under low sampling budgets.
- JHU: A Two-fold Randomization Framework for Impulse Control Problems** Dec 2024-Sep 2025
- Advisor(s): Haoyang Cao.
 - Impulse controls are typically used for modelling exchange rate, transaction cost, insurance, and cash management, where the controller cannot continuously adjust the system but instead intervenes at discrete optimal times.
 - Introduced a two-fold randomization scheme over intervention time and jump size to enable exploration of potentially unknown environment.
 - Designed an offline, model-free continuous-time RL algorithm and proved the policy improvement theorem.
 - Derived compound-Poisson and randomized optimal-stopping formulations leading to the same HJB equation.
 - Proved existence, uniqueness and regularity of the solution to randomized framework, and a convergence theorem between the randomized framework and its classical counterpart.
- JHU: Training-free versus Training-based Intent Classification in LLMs: Accuracy, Robustness, and Failure Modes** Aug 2025-Sep 2025
- Advisor(s): Soufiane Hayou.
 - Studied intent classification for LLMs, which categorizes user prompts into predefined classes
 - Proposed two lightweight training-free intent classification methods. Both methods operate on statistics of LLM prefill-phase feature vector. Developed theory explaining their respective strengths.
 - Conducted a systematic empirical study comparing training-free and training-based intent classification across seven LLMs (1B--32B parameters). Both paradigms saturate on easy tasks, while training-based methods typically outperform training-free ones on harder, fine-grained distinctions. Meanwhile, training-free methods are typically more robust to mixed-intent prompts and adversarial rephrasings, making them potentially attractive for practical LLM routing settings where inputs are often noisy or ambiguous.

NUS: Memory-Efficient Gradient Unrolling for Large-Scale Bilevel Optimization

March 2024-May 2024

- Advisor(s): Kenji Kawaguchi.
- Proposed (FG)²U, a Hessian-free forward-gradient unrolling method that produces an unbiased stochastic estimate of the bilevel meta-gradient without storing the full inner-optimization trajectory.
- Conducted analysis on variance and convergence of (FG)²U and to a parallelizable two-phase optimization strategy that combines efficient approximate updates with accurate late-stage meta-gradient refinement.
- Evaluated the method on image data condensation, online adaptation of language models, and black-box PDE discovery; demonstrated scalability to GPT-2 XL with 1.5B parameters and substantially longer unrolling depths.

Caltech: Distributionally Robust Policy Gradient for Offline Contextual Bandits

Mar 2022-Oct 2022

- Advisor(s): Anima Anandkumar from Caltech, Pan Xu from Duke University.
- Developed a minimax distributionally robust policy-gradient method to address logging-to-target distribution shift in offline contextual bandits.
- Established generalization and convergence guarantees and extended the framework to settings with limited online exploration.
- Implemented experiments on MNIST and UCI Optdigits to evaluate robustness under policy and distribution shift.

WORKING AND TEACHING EXPERIENCE

- Teaching Assistant at IBM's Thomas J. Watson Research Center July 2026
Brief Description: I served as a teaching assistant for the **Dynamical Systems for Machine Learning and AI** graduate summer workshop, where I prepared lecture notes, delivered several tutorial sessions, and mentored students on their projects.
- Instructor of Probability Theory at AMS Master Student Orientation Aug 2025
- Teaching Assistant for Introduction to Computational Mathematics Aug 2024-May 2025

HONORS AND AWARDS

- Duncan Research Fund at Johns Hopkins University June 2026
- Travel Award for Poster Presentation at ECMF9 Sep 2025
- Outstanding students Scholarship of Shanghai Jiao Tong University 2019-2023 (four times)
- Honors scholarship for students of Zhiyuan College 2019-2023 (four times)

PUBLICATIONS

In Preparation

1. Haoyang Cao*, **Zhouhao Yang***, "Mean Field Games for Randomized Impulse Control." In preparation (2026).

Under Review / In Revision

2. **Zhouhao Yang**, Yezhen Wang, Kenji Kawaguchi, Vladimir Braverman, Haoyang Cao, "Heavy-Tailed Flow Matching via Random Clocks." Under review in Neurips 2026.

Published / To Appear

3. **Zhouhao Yang***, Nan Chen*, Soufiane Hayou, "Training-free versus Training-based Intent Classification in LLMs: Accuracy, Robustness, and Failure Modes." To appear in COLM 2026.
4. Haoyang Cao*, Yuchao Dong*, **Zhouhao Yang***, "A Two-fold Randomization Framework for Impulse Control Problems." To appear in Siam Journal on Control and Optimization, 2026.
5. Yan Hao Ling, **Zhouhao Yang**, Jonathan Scarlett, "Statistical Mean Estimation with Coded Relayed Observations," To appear in IEEE Transactions on Information Theory, 2025.
6. Zheyuan Hu*, **Zhouhao Yang***, Yezhen Wang*, George Em Karniadakis, Kenji Kawaguchi, "Bias-variance trade-off in physics-informed neural networks with randomized smoothing for high-dimensional PDEs." SIAM Journal on Scientific Computing 47.4 (2025): C846-C872.
7. Qianli Shen, Yezhen Wang, **Zhouhao Yang**, Xiang Li, Haonan Wang, Yang Zhang, Jonathan Scarlett, Zhanxing Zhu, Kenji Kawaguchi, "Memory-Efficient Gradient Unrolling for Large-Scale Bi-level Optimization," The Thirty-eighth Annual Conference on Neural Information Processing Systems. 2024.
8. **Zhouhao Yang***, Yihong Guo*, Pan Xu, Anqi Liu, and Anima Anandkumar, "Distributionally robust policy gradient for offline contextual bandits," in International Conference on Artificial Intelligence and Statistics, pages 6443–6462. PMLR, 2023.
9. **Zhouhao Yang**, Xingyu Xu and Yuantao Gu, "A General Framework for Accurate and Private Mean Estimation," in IEEE Signal Processing Letters, vol. 29, pp. 2293-2297, 2022, doi: 10.1109/LSP.2022.3219356.

SKILLS

- Python (PyTorch, JAX, pandas, numpy, scipy, etc.), MATLAB, Linux, git